

SOC WEEK-1 REPORT

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Title: Networking and Security Operations with SIEM, Forensics, and Traffic Analysis

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Introduction

This report summarizes the Week-1 tasks performed during the SOC Internship training. The objective of this week was to gain foundational theoretical knowledge of SOC operations and perform basic hands-on tasks such as log analysis, monitoring, Snort rule testing, and incident documentation.

All practical activities were performed in a virtual lab setup using:

Host Machine: Windows 10

VM-1: Kali Linux — 192.168.1.90

VM-2: Metasploitable — 192.168.1.100

1. SOC Fundamentals and Operations

1.1 Purpose of a SOC

A Security Operations Center (SOC) monitors, detects, investigates, and responds to security events.

Key responsibilities include:

- Continuous monitoring of systems and logs
- Alert analysis and triage
- Threat detection and hunting
- Incident response and reporting

1.2 SOC Roles

Tier 1 Analyst: Initial alert monitoring & triage

Tier 2 Analyst: Investigation & correlation

Tier 3 Analyst: Threat hunting & advanced analysis

SOC Manager: Oversight, workflow, reporting

1.3 Frameworks Covered

NIST Cybersecurity Framework

MITRE ATT&CK (tactics & techniques)

NIST SP 800-61 (Incident Handling Guide)

2. Security Monitoring Basics

Security monitoring aims to detect suspicious behavior, policy violations, and potential attacks.

2.1 Tools Studied

Wireshark — Packet capture and analysis
Elastic SIEM — Log visualization and alerts
Syslog — Centralized log forwarding

2.2 Key Concepts Learned

- False Positives / False Negatives
- MTTD (Mean Time to Detect)
- MTTR (Mean Time to Respond)

2.3 Practical Exposure

Applied Wireshark filters to inspect traffic
Observed TCP handshake, resets, and anomalies
Basic SIEM dashboards reviewed

3. Log Management Fundamentals

Log management includes log collection, parsing, normalization, storage, and searching.

3.1 Log Types Studied

Windows Event Logs
Syslog (Linux)
Application Logs (Apache HTTP logs)

3.2 Hands-On Activities

- Used logger "Test Message" on Kali to generate syslogs
- Explored /var/log/auth.log and /var/log/syslog
- Reviewed failed SSH login attempts

3.3 Practical Task — Event ID 4625

Analyzed Windows failed login (Event ID 4625) using Event Viewer:
Detected brute-force patterns
Exported logs to CSV

Identified source account, timestamp, and failure reason

4. Security Tools Overview

4.1 Tools Covered

SIEM: Elastic Stack
IDS/IPS: Snort
Vulnerability Scanner: Nessus Essentials
Endpoint Visibility: Osquery

4.2 Practical Task — Snort Rule

Created a custom Snort rule on Kali:

```
alert tcp any any -> any 80 (msg:"Malicious Domain"; content:"malicious.com"; http_uri;  
sid:1000001;)
```

5. Basic Security Concepts

Covered Concepts

- CIA Triad: Confidentiality, Integrity, Availability
 - Defense-in-Depth
 - Zero Trust Model
 - Difference between Threat, Vulnerability, and Risk
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6. Security Operations Workflow

SOC workflow stages covered:

1. Detection — Identifying alerts
 2. Triage — Validating & prioritizing alerts
 3. Investigation — Deep log review & IOC correlation
 4. Response — Containment, mitigation
 5. Documentation
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Activities Performed:

Basic triage flow practiced

Investigated test failed login event

Mapped activity to MITRE ATT&CK Tactic: Credential Access (TA0006)

7. Incident Response Basics

NIST IR Lifecycle Studied

1. Preparation
2. Identification
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned

Completed a basic tabletop simulation:

Simulated ransomware alert investigation

Prepared a mock response sequence

8. Documentation Standards

Practiced documentation including:

- Incident Report
- Analyst Notes
- Triage Summary
- Post-Incident Summary

Template fields used:

- Date & Time
 - Source IP
 - Event ID
 - Description
 - Analyst Action
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PRACTICAL APPLICATION SUMMARY

1. Log Analysis Practice

Activities Completed:

- Analyzed Windows 4625 events
- Generated test failed logins (wrong password attempts)
- Parsed Chrome history using Zimmerman's LECmd tool
- Exported logs to CSV

2. Security Event Documentation

Date/Time: 2025-11-14

Source IP: 192.168.1.100

Event: Multiple failed login events (Event ID 4625)

Severity: Medium

Analyst Action:

- Investigated login attempts
 - Identified brute-force behavior
 - Locked the affected account
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- Documented findings in SOC logbook

3. Monitoring Dashboards

- Configured and reviewed dashboards in Elastic SIEM:
- Top failed login sources
- Event ID frequency graphs
- Alerts by severity

4. Alert Rule Configuration

Created SIEM rule:

Condition:

More than 5 failed logins within 5 minutes

Testing:

- Attempted repeated SSH failures from Kali
 - Alert triggered in SIEM
 - Validated effectiveness
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Learning Summary

Week-1 provided strong foundational knowledge of:

SOC structure and workflows
SIEM operations
Log analysis
Basic incident detection
Snort rule creation
Security documentation methods
